Midterm Project-Deep Learning Report

By: Zachary Thornton

Problem definition and Dataset curation:

The purpose of this code is to solve a multi-class image classification problem using deep learning. In this program, its task is to classify/identify images of handwritten digits into one of ten categories (0-9). When it's given a 28 x 28 image of a handwritten digit, the model predicts what digit is being shown in the image. This is considered a supervised learning problem because each image being put into the program is matched up with a known label during the training. The model learns to map the input (images) into their corresponding digit labels 0-9. The dataset being used in project is the MNIST dataset. This dataset contains 70,000 grayscale images of handwritten digits. 60,000 of the images are used for training and 10,000 images are used for testing. Each image is 28 x 28 pixels, and it contains one handwritten digit between 0-9.

Deep Learning models:

The model that is used in this project is a Convolutional Neural Network (CNN). CNNs are normally known to be very good for image-classification tasks. The reason for this is because they automatically learn different spatial features like edges, patterns, and shapes. The model looks at small sections of the image at a time, and it builds an understanding of what the digit looks like. When the image goes through the different layers, the model gets out more detailed features, like recognizing simple edges on the digit versus seeing more complex shapes that could represent specific numbers. After the image has gone through all the layers in the program, the model decides which digit the image most likely is. During training, the model compares it's predictions to the right answers and it adjusts itself to improve the accuracy.

The Results:

```
(.venv) zacharythornton@Zacharys-Air AI_100_Midterm % python3 mnist_cnn.py --epochs 3
Using device: mps
Epoch 01/3 | loss=0.1506 | val_acc=98.40%
  Saved new best model -> mnist_cnn_best.pt (val_acc=98.40%)
Epoch 02/3 | loss=0.0437 | val_acc=98.93%
  Saved new best model -> mnist_cnn_best.pt (val_acc=98.93%)
Epoch 03/3 | loss=0.0337 | val_acc=98.93%
Final TEST accuracy: 99.15%
(.venv) zacharythornton@Zacharys-Air AI_100_Midterm %
```

This is the output from the program above. As you can see during the training, the loss was steadily decreasing between the different epochs. The significance of this is that it's showing that the model is learning more and more patterns from the data and becoming smarter. After training, the model got a final test accuracy of 99.15%, which means that the model correctly classified around 99 out of every 100 digits in the test data inside of the dataset. It's a good thing that the validation accuracy and test accuracy were alike, because this suggests that the model has a good generalization to new data. This means that the model isn't learning training data too closely and capturing random details and patterns in the photos instead of focusing on the correct task of what to look for in the digit images.

Lessons & Experience I have learned:

During this project, I learned a lot more about how deep learning models learn from data and make predictions, especially about how Convolutional Neural Networks work and process image data. I found it very interesting to learn what the different parts of my program represented. I also learned more about the importance of validation datasets and how their used to make sure the model generalizes okay to new data. This tied in with me learning more on why it's important to use the same model with different data to make sure it performs okay with unseen data. I ran into a lot of technical challenges in implementing my program and getting it to run, and even though working through these different issues were annoying at times, it helped me learn more about managing programming environments and debugging. Overall, this project strengthened my understanding of machine learning concepts and gave me more experience with applying them.